Lecture 14

FAIR and Open Science

Principles, Reproducibility, and Open Research

Computing for Data-Driven Biology · 5BI00A
Master's Programme in Bioinformatics · Umeå University

Session Aims & ILO Links

1	State and explain the four FAIR principles with a concrete bioinformatics example for each	ILO 3
2	Describe the open science landscape — open access, open data, open source, open peer review	ILO 3
3	Explain why Git is reproducibility infrastructure, not just version control	ILO 3, 12
4	Describe FAIR4RS and explain how it extends FAIR to software and code	ILO 3
5	Connect open science practices to the UN Sustainable Development Goals	ILO 3
6	Identify the criteria for the essay assessment and describe what VG-level work looks like	ILO 13

Opening Poll

What does FAIR stand for?

Word cloud — type the four words (one at a time)

Also: Have you ever tried to reanalyse or reproduce published bioinformatics data?

- Yes — successfully
- Yes — but hit problems
- No — tried to find data
- No — never attempted

And: word cloud — in one word, why does reproducibility matter in science?

Part 1

The reproducibility problem — and why it matters in bioinformatics

The Reproducibility Problem: Evidence

This is not a fringe concern — it is a structural problem with measurable consequences.

70%

of researchers have failed to reproduce another scientist's experiment

Baker, Nature (2016), n=1,576

50%

have failed to reproduce their own experiments

Baker, Nature (2016)

<50%

of high-profile RNA-seq papers were computationally reproducible in a 2022 systematic study

Wijesooriya et al. (2022)

Consequences in bioinformatics:

Wasted public funding · Clinical decisions based on unreproducible results · Literature pollution through citations of flawed work · Career consequences for the researchers involved

The Causes — and Their Solutions

Irreproducibility in bioinformatics is largely preventable. Each cause has a direct solution.

✗ Undescribed software versions	✓ Pin exact versions in methods; use containers
✗ Unavailable raw data	✓ Deposit in ENA/SRA/GEO with a persistent accession
✗ Undocumented parameters	✓ Specify all non-default parameters; commit scripts to Git
✗ Code never version-controlled	✓ Git from day one; tag commits that generate manuscript results
✗ Statistical reporting without pipeline	✓ Provide analysis code; workflow manager (Snakemake/Nextflow)
✗ Format and dependency issues	✓ Containers (Docker/Singularity) package code + environment together

Part 2

The FAIR principles — what they mean in practice

The FAIR Principles: An Overview

Wilkinson et al. (2016) Scientific Data 3:160018 · DOI: 10.1038/sdata.2016.18

FAIR is primarily about enabling machines to find, access, integrate, and reuse data — not just humans.

F

Findable

Persistent identifiers · Rich metadata · Indexed in searchable resources

A

Accessible

Standardised retrieval protocol · Authentication where needed · Metadata persists even if data is removed

I

Interoperable

Standard formats · Ontology-based metadata · Cross-references to related resources

R

Reusable

Rich metadata · Data licence · Provenance · Community standards (MINSEQE, MIxS)

FAIR ⁄ open. Controlled-access data can be fully FAIR. The opposite of FAIR is not 'restricted' — it is 'unusable by machines.'

Findable and Accessible: In Practice

F

Findable

Persistent identifiers (DOIs, accession numbers, ORCIDs) · Rich metadata indexed in searchable databases · The same dataset accessible from ENA, GEO, or ArrayExpress via a stable accession

✓ Raw reads in ENA with accession PRJEB12345, samples annotated with EFO ontology terms
✗ Data in supplementary Excel file — no identifier, not indexed, lost if publisher changes hosting

A

Accessible

Retrievable by identifier via standardised protocol (HTTPS, FTP) · Authentication documented where needed · Metadata persists as a tombstone record even if data must be withdrawn

✓ GTEx: raw data via documented dbGaP application; processed summaries fully open via stable URL
✗ 'Data available upon reasonable request' — corresponding author has since left academia

Interoperable and Reusable: In Practice

I

Interoperable

Standard formats (FASTQ, BAM, VCF, GFF) · Ontology-based metadata (GO, EFO, NCBI Taxonomy, UBERON) · Cross-references to related datasets · APIs with documented endpoints

✓ Sample annotated as UBERON:0002107 (liver) + EFO:0001272 (treated) — machine-readable, comparable across studies
✗ Sample annotated as 'liver (adult, treated with drug X)' — unstructured, cannot be automatically integrated

R

Reusable

Rich metadata sufficient to assign samples without author contact · Data licence (CC-BY, CC0) · Full provenance chain · Software version pinning · Community metadata standard declared

✓ Sample table with taxonomy ID, tissue ontology, treatment, time point; CC-BY licence; software versions specified; Snakemake workflow in Zenodo with DOI
✗ Samples labelled 'control' and 'treated'; no licence; 'standard parameters used'; code available upon request

Preview — Jamie McCann joins us later in this session to introduce PlantGenIE; you will query its REST API endpoints yourself in the self-study exercises that follow.

💬 *Pair Discussion — 4 minutes*

Think of a bioinformatics paper you have read — or one from your undergraduate degree.

1. Was the raw data deposited somewhere you could find with a persistent identifier?
2. Could you have told which samples were which from the metadata alone?
3. Was the analysis code available — and if so, were software versions specified?
4. Could you reproduce the main result from what was published?

Share back: what was the most common failure mode in your examples?

Quick Check — Before the Break

Q1. A researcher says data is 'available upon reasonable request'. Which principle does this most directly violate?

- A) Findable — no identifier
- B) Accessible — no standardised retrieval protocol ✓
- C) Interoperable — format not specified
- D) Reusable — no licence

Q2. Dataset A uses 'liver, treated.' Dataset B uses UBERON:0002107 + EFO:0001272. Which is more interoperable?

- A) Dataset A — more human-readable
- B) Dataset B — ontology terms are machine-readable and enable automated integration ✓
- C) Equally interoperable — same information
- D) Neither — interoperability needs a specific file format, not metadata

Q3. A researcher archives code on GitHub. Is this sufficient for FAIR4RS?

- A) Yes — code is publicly accessible
- B) No — GitHub repos can be deleted; a persistent DOI via Zenodo is required ✓
- C) No — GitHub doesn't support bioinformatics
- D) Yes, if the README describes the software

☕ Break — 10 minutes

Back at: ____________

After the break: open science · Git as reproducibility · FAIR4RS · SDGs · essay assessment

Part 3

Open science · Git as reproducibility · FAIR4RS · Essay

The Open Science Landscape

📄 Open Access

Gold OA: publisher makes final version immediately open (often APC required)
Green OA: author self-archives in repository (PubMed Central, bioRxiv)
Diamond OA: no cost to author or reader
Preprints: bioRxiv/medRxiv — rapid sharing before peer review

Vetenskapsrådet and ERC mandate OA for funded research.

📊 Open Data

Data in public repositories with persistent identifiers and rich metadata
General repositories: Zenodo, Figshare
Domain-specific: ENA, GEO, PRIDE, ArrayExpress
FAIR principles apply directly here
Data management plans now required by most funders

💻 Open Source Software

Source code available under an open licence (MIT, Apache 2.0, GPL)
Tools you already use: BLAST+, samtools, bcftools, bedtools — all open source
Licences matter: without one, code cannot legally be reused
Resource: choosealicense.com

🔍 Open Peer Review

Reviewer identities and/or review content made public
Increases transparency and accountability
Adopted by eLife, PeerJ, F1000Research, and others
Preprint review platforms: Review Commons, PREreview

The Turing Way: https://the-turing-way.netlify.app/ · FAIRsharing: https://fairsharing.org/ · Plan S: https://coalition-s.org/

Git as Reproducibility Infrastructure

You have been using Git for version control. It is also — when used well — the backbone of reproducible science.

📌 **Every commit is a reproducibility checkpoint**

The complete history of every analysis file is preserved. When a reviewer asks 'exactly what did you do?', you can show the exact code at the time of submission.

🏷️ **Tags connect commits to publications**

Create a tag (e.g. v1.0-paper-submission) at the commit used to generate manuscript results. Reference this tag in the methods section. Readers can retrieve that exact version.

🌐 **Archive repositories beyond GitHub**

GitHub repositories can be deleted. For publications, archive via Zenodo (integrates directly with GitHub — each tagged release gets a DOI automatically) or Software Heritage.

📦 **Containers extend reproducibility to the environment**

Code alone is insufficient — the software environment matters too. Docker or Singularity containers package code + all dependencies together. This is FAIR4RS in practice.

Reproducible Reports: Notebooks and R Markdown

Jupyter Notebooks

Code cells, live output, and narrative Markdown cells in one .ipynb file · re-executed top to bottom to regenerate every output

✓ Used in Lecture 15 to build a minimal example from this session's own results, and in depth in Python Programming for Bioinformatics

R Markdown + knitr

Interleaves R code chunks with narrative Markdown text in one .Rmd file · knitr renders it to HTML, PDF, or Word

✓ Rendering the same .Rmd file always regenerates the same report. Used in depth in Data Science for Biology with R

Not cosmetic: a reader sees the reasoning alongside the code and the result it produced — not a script to run blind and outputs to reconstruct from memory.

Writing Reproducible Methods

A Methods section is a specification precise enough that someone else could redo the analysis without a single follow-up question.

🔢 **Exact software versions**

"We used BLAST" is not reproducible. State the tool and exact version, e.g. blastp v2.17.0 (BLAST+) — versions can give different results

📖 **A citation for every tool**

Not just a version number — citing the software credits the people who built and maintain it, the same courtesy as citing a paper or dataset

⚙️ **Every parameter, including defaults**

Default parameters change between versions. Stating the exact version pins what those defaults were; any non-default parameter must be stated

🧬 **Database and reference genome versions**

"The human reference genome" is not reproducible. State the exact assembly and release, e.g. GRCh38.p14 (Ensembl release 110), with a citation

Worked Example: Not Reproducible vs Reproducible

The same standard applies whether you are writing this Methods section, or assessing someone else's for the essay.

Not reproducible	Reproducible
Reads were aligned to the reference genome with default parameters.	Reads were aligned to GRCh38.p14 (Ensembl release 110) with STAR v2.7.11a (Dobin et al., 2013, Bioinformatics 29(1): 15–21), using --outFilterMismatchNmax 2; all other parameters were left at their v2.7.11a defaults.
Differential expression was assessed using standard statistical methods.	Differential expression was assessed with DESeq2 v1.42.0 (Love et al., 2014, Genome Biology 15: 550) in R v4.3.2, using the default Wald test and an adjusted p-value threshold of 0.05 (Benjamini-Hochberg correction).

The Gold Standard: Data + Code, Together

The inputs

Raw data deposited at a persistent repository — Zenodo, Figshare, or a domain archive (ENA, GEO, PRIDE)

✗ Data without code tells a reader what went in, but not how the reported values came out

The producers

Analysis code deposited at a version-controlled repository, ideally a tagged release archived to Zenodo with its own DOI

✗ Code without data is unusable — a script cannot run against nothing

Together, and only together, they make every table, figure, and stated value something a reader can regenerate, not merely take on trust.

AI-Assisted Reproduction: A Live Question

Verifying full reproducibility used to mean a human manually rerunning a pipeline — slow, expensive, and rarely done. That is changing.

Partial success

A multi-agent LLM system attempted to independently regenerate results from a published scRNA-seq study — real progress, with documented failure modes

Bersenev et al., bioRxiv (2024). DOI: 10.1101/2024.04.08.588614

Mixed evidence

In a large controlled study of AI-assisted reproduction of social-science findings, AI-assisted teams matched human-only teams — but fully autonomous, AI-led attempts performed markedly worse

Brodeur et al., PNAS (2026). DOI: 10.1073/pnas.2524747123

The test to apply

Could a stranger, or an AI system with no access to you, regenerate this figure from what you have deposited? Treat that as a genuine test of your own work before submission.

FAIR for Research Software (FAIR4RS)

Chue Hong et al. (2022) · DOI: 10.15497/RDA00068

The same four principles — applied to software and code.

F	**Findable** ✗ Script emailed as attachment on request, no identifier	✓ GitHub repo with Zenodo DOI, registered in bio.tools
A	**Accessible** ✗ Compiled binary only, no source code	✓ Source code on GitHub with stable URL; Zenodo archive for permanence
I	**Interoperable** ✗ Reads and writes proprietary formats, no documented API	✓ Standard I/O formats (FASTA, VCF), documented CLI and API
R	**Reusable** ✗ No licence, no README, no version history, ad hoc variable names	✓ MIT licence, README with install instructions, pinned dependencies, version tags

Open Science and the UN Sustainable Development Goals

Open science is not merely good academic practice — it is an instrument of global equity and environmental sustainability.

SDG 3

Good Health & Well-Being

Open access to genomic, clinical, and pharmacogenomic data accelerates medical research and enables researchers in low- and middle-income countries to participate in global science.

SDG 14

Life Below Water

eDNA metabarcoding data deposited in ENA using SILVA and BOLD reference databases enables global marine biodiversity monitoring. Open data is the raw material of conservation policy.

SDG 15

Life on Land

GBIF, BOLD, SILVA, and UNITE are open data infrastructures directly used by CBD monitoring frameworks. Biodiversity assessment at scale is only possible because these databases are open and FAIR.

SDG 17

Partnerships for the Goals

Open science is itself a mechanism of international partnership — open data, open access, and open software enable researchers everywhere to build on shared knowledge, regardless of institutional resources.

Guest Contribution

Jamie McCann introduces PlantGenIE

PlantGenIE: Research Infrastructure at Umeå

🌱 *Developed at Umeå University · plantgenie.se · github.com/plantgenie*

🌲 Genome browsers

Reference genomes for *Picea abies* (Norway spruce) and *Populus tremula* (European aspen), with gene models, annotations, and synteny tools

📈 Expression data

Integrated RNA-seq expression atlases across tissues and conditions, with co-expression network tools for identifying gene modules

🔍 BLAST and search

Sequence similarity search against conifer and aspen genomes — accessible via web interface and API

⚡ REST API

All data accessible programmatically via documented HTTP endpoints returning JSON and FASTA. You will use this directly in the self-study exercises in Lecture 15.

Jamie McCann will now introduce PlantGenIE and explain the API design decisions that make it Interoperable.

Essay Assessment

Introducing the FAIR compliance essay

The FAIR Essay: What It Asks

Full details, marking criteria, deadline, and submission instructions are in Canvas.

The task

Critically evaluate the FAIR compliance of a provided published bioinformatics study and its associated data. Your evaluation should cover all four FAIR principles, the sufficiency of metadata for reuse, and the reproducibility of the computational analysis. Make specific, evidenced recommendations.

G

Pass

Correctly identifies and describes FAIR issues · Applies principles accurately · Notes data deposition and code availability · Observations are relevant and accurate

VG

Distinction

All of the above, plus: Evaluates consequences of each FAIR issue · Makes specific, concrete recommendations · Distinguishes FAIR spirit from letter · Structured, evidenced, analytical — not merely descriptive

What to Look For: The Assessment Checklist

F

- ☐ Accession number in the paper? Is it persistent (SRA/ENA/GEO/DOI) or just a URL?
- ☐ Can you find the dataset via a database search engine without knowing the accession?

A

- ☐ Can you actually download the data via HTTPS or FTP?
- ☐ If access-controlled, is the application process documented?

What to Look For: The Assessment Checklist (continued)

I

- ☐ Are data in standard formats (FASTQ, BAM, VCF)?
- ☐ Are samples annotated with ontology terms or free text?
- ☐ Is the reference genome version specified with a versioned identifier?

R

- ☐ Is there sufficient metadata to assign samples to conditions without contacting the authors?
- ☐ Is there a data licence? Are software versions specified for all tools?
- ☐ Is analysis code available, documented, and versioned?
- ☐ Is a community metadata standard declared (MINSEQE, MIxS)?

What a Strong Response Looks Like

G level

"The raw data was deposited in the NCBI SRA with accession SRP123456 (F). The data can be downloaded (A). The authors used standard FASTQ format (I). The methods section describes the analysis (R)."

Describes facts correctly but does not evaluate quality, consequences, or what could have been improved.

VG level

"While raw reads were deposited in SRA (SRP123456), Findability is compromised: no EFO annotations were deposited for the six conditions, making semantic discovery impossible without prior knowledge of the accession. Regarding Reusability: HISAT2 v2.1.0 and DESeq2 v1.20.0 are specified, but exact parameters are omitted and no code is available — 'standard parameters were used' is insufficient for reproduction. This is remedied straightforwardly by depositing a MINSEQE-compliant sample table and committing analysis scripts to a versioned repository at submission."

What You Should Know After This Session

- ✓ State all four FAIR principles and give a concrete bioinformatics example for each
- ✓ Explain the difference between accessible and open — controlled access can be FAIR
- ✓ Explain why ontology terms make metadata more interoperable than free-text annotations
- ✓ Describe three components of open science and name one key resource for each
- ✓ Explain how Git functions as reproducibility infrastructure beyond version control
- ✓ Describe FAIR4RS and give an example of the difference between FAIR and non-FAIR software
- ✓ Apply the FAIR framework to a bioinformatics study, identifying issues and their consequences
- ✓ Describe what distinguishes a VG-level essay response from a G-level one

Next: self-study exercises in FAIR in Practice — working with the PlantGenIE API, a GEO/ENA metadata assessment, and a paper reproducibility exercise.